

23. Answer: D

Using $F_B = Bqv_{\perp} = Bqv \sin \theta$,

$$v = \frac{F_B}{Bq \sin \theta}$$

$$= \frac{4.3 \times 10^{-14}}{0.088 \times 1.60 \times 10^{-19} \times \sin 20^\circ}$$

$$= 8.9 \times 10^6 \text{ m s}^{-1}$$